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When was the U.S. housing downturn predictable? A comparison of univariate forecasting methods

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ABSTRACT

This paper uses three classes of univariate time series techniques (ARIMA type models, switching regression models, and state-space/structural time series models) to forecast, on an ex post basis, the downturn in U.S. housing prices starting around 2006. The performance of the techniques is compared within each class and across classes by out-of-sample forecasts for a number of different forecast points prior to and during the downturn. Most forecasting models are able to predict a downturn in future home prices by mid 2006. Some state-space models can predict an impending downturn as early as June 2005. State-space/structural time series models tend to produce the most accurate forecasts, although they are not necessarily the models with the best in-sample fit.

1. Introduction

The purpose of this study is to compare the ability of a variety of univariate time series forecasting techniques to predict the downturn in the U.S. housing market starting around 2006. Identifying a relatively simple-to-implement method that could have predicted early on a significant downturn in housing prices may be helpful for future investors in the housing market and policy makers alike.

Univariate forecasting techniques are of practical relevance because no costly data collection is involved, no potentially controversial economic theory or extensive empirical model building is required, and the forecasting can be done on a frequent and routine basis. In fact, the majority the models analyzed in this study can be run in a close to automated mode, which increases the chances that they are utilized. We consider in this study three classes of univariate models, (a) Box-Jenkins style methods, (b) regime-switching models, and (c) state-space models. In the latter class of models we focus on standard unobserved component or structural time series models as popularized in economics by Harvey (1989) and more recently extended for forecasting by others (Hyndman, Koehler, Ord, & Snyder, 2008).

ARIMA-style models turn out to fare rather well in predicting prices before the housing downturn, even better than regime-switching models, which, however, seem to fit better in-sample. Examples of studies using these methods are provided by Crawford and Fratantoni (2003), Miles (2008), and D'Agostino, McQuinn, and O'Reilly (2008). More recent evidence by Larson (2011) suggests that simple univariate model versions of the ARIMA or state space type have trouble predicting the housing market downturn. Theory driven multivariate models, in particular vector error-correction models, appear to perform better.

Numerous other studies also follow the theory driven path and try to tie prices to economic fundamentals. Examples are Campbell, Davis, Gallin, and Martin (2009), Gallin (2006, 2008) and Rapach and Strauss (2009). But, as pointed out by Rapach and Strauss (2009), fundamentals seem to work well in forecasting prices only in situations where prices are not moving much, such as in the interior states of the U.S. prior to 2006, not in the coastal states

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which experienced strong housing price inflations over the same time period.

The importance of economic theory is also highlighted by Gupta, Kabundi, and Miller (2011), who compare the forecasts of the micro-based 10-variable dynamic stochastic general equilibrium (DSGE) model of Iacoviello and Neri (2010) to a number of multivariate but atheoretical models of the VAR and Bayesian VAR type. They find that only the theory-based DSGE can predict the turning point in housing prices in 2006. Other work by the same authors, Gupta and Das (2010) and Das, Gupta, and Kabundi (2011), confirms that models with many variables tend to predict better than models with fewer variables, but they also suggest that the identification of turning points is rather tenuous.

Although the preponderance of the evidence seems to favor more involved theory-driven models, recent results by Huang (2012) on regime switching models imply that simpler approaches should not be counted out too early. The evidence provided in this study confirms that univariate models, in particular state space models of the structural time series type, have a useful role to play in forecasting turning points. The best performing models are able to predict a downturn in housing prices as early as the middle of 2005.

2. Data and methodology

All our models make use of the Case-Shiller 10-market composite home price index, which is widely used as a house price indicator and easily available over a long time horizon.[1] The series we employ is recorded at a monthly frequency and seasonally not adjusted. Seasonal variation is taken into account by including either seasonal dummy variables, defined so as to sum to zero, or a combination of sine/cosine variables. The latter are defined as $sin\,.\,t = \sin(2\pi t/12)$ and $cos\,.\,t = \cos(2\pi t/12)$, where t represents a time trend.

We provide 6-, 12-, and 18-months-ahead dynamic out-of-sample forecasts at 6 different forecast points, all 6 months apart and starting in December of 2004. The forecasts are based on models estimated on monthly data, starting with October 1987 as the initial point of the sample and ending with the forecast points.

For the purpose of estimating and forecasting the models, we convert the monthly price index to a monthly return series by taking the monthly log difference and then rescale this return series by multiplying by 100. The return series can be interpreted as the monthly growth rate of the housing price index or, more simply, as the monthly rate of house price inflation. For the graphical illustration of the forecasts, the forecasts of the return series are translated back into changes of the house price index.

The forecasts draw on three classes of univariate models, (a) Box-Jenkins style ARIMA models, (b) regime-switching models, and (c) structural time series models of the type introduced into economics by Harvey (1989) and further developed for forecasting purposes by Hyndman et al. (2008). The third class of models is also known as unobserved component or state-space models. For the purpose of comparison, we add a *naive* forecasting model, which predicts next period's value to be identical to this period's value, and a neural network model, which uses a large number of parameters for the model fit.[2]

[1] The index can be obtained from: http://www.standardandpoors.com/indices/sp-case-shiller-home-price-indices/en/us/?indexId=spusa-cashpidff–p-us—-

[2] A convenient discussion of the neural network approach to modeling return series is provided in Franses and van Dijk (2000, chapter 5).

2.1. Box Jenkins style models

In this model class we consider classical Box-Jenkins style ARIMA models and fractionally integrated ARMA (ARFIMA) models. These types of models can be written in the format

$$\phi(L)\Phi(L^s)(1-L)^d y_t = \theta(L)\Theta(L^s)\varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2), \tag{1}$$

where $\phi(L)$ represents non-seasonal autoregressive lags, $\Phi(L^s)$ seasonal autoregressive lags with $s = 12$ in our case; exponent d stands for standard or fractional differences[3]; $\theta(L)$ identifies non-seasonal moving average terms and $\Theta(L^s)$ seasonal moving average terms. Seasonal differencing is not needed for the given data set.

One of the modeling aspects of this class of models is pretesting for the stationarity in the mean and in the season.[4] Financial return series are typically assumed to be stationary. The return on house prices is a borderline case. By selecting different sub-samples and different stationarity or non-stationarity tests, it is possible to find empirical evidence in favor of both stationarity and non-stationarity. Given this ambiguity, we estimate models with and without first differencing and models with fractional differencing. Models are chosen based on AIC criteria.

To optimize the model fit and also to make the results replicable, we use the automated ARIMA fitting procedure in the *R* package *forecast*.[5] Fractionally differenced ARFIMA models are estimated using the *arfima* function in the *R* package *forecast*. This function uses a step-wise process that utilizes automated ARIMA fitting along the lines of Hyndman and Khandakar (2008) combined with the Haslett and Raftery (1989) algorithm to estimate the value of the differencing parameter *d*.[6]

2.2. Regime switching models

We consider three types of models: Markov switching, smooth transition regime switching, and regime switching based on nonlinear moving average models.[7] In the first model type, a regime switch follows a Markov process that is defined by transition probabilities. The regime that is active at time t is not tied to any observable variable. In the latter two models the regime switch typically depends on past observed values of the variable to forecast.

Based on some preliminary tests to achieve convergence, we estimate a Markov switching model of the form

$$(1-\rho_1 L-\rho_2 L^2)(y_t-\phi_{0,s_t}-\phi_{1,s_t}t-\phi_{2,s_t}\sin.t-\phi_{3,s_t}\cos.t)=\varepsilon_t, \tag{2}$$

where $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ and where s_t identifies the state or regime present at time t. The autoregressive parameters ρ_1 and ρ_2 and

[3] $d = 0$ for stationary y_t, $d = 1$ for y_t following an $I(1)$ process, and $|d| < 0.5$ for fractionally integrated y_t.

[4] We ignore in this paper issues of variation in the variance as one can reasonably argue that conditional volatility is not a key feature of pricing in the housing market.

[5] The function *auto.arima* in version 4.06 of the package *forecast* draws on the *arima* function of the standard *stats* package of *R* and selects the best model based on the AIC criterion, with the maximum order of AR and MA terms set to 5 and those for seasonal terms set to 2. The order of first differencing is based on the KPSS test. More detail is provided in Hyndman and Khandakar (2008).

[6] First, the differencing parameter *d* is determined based on a model with 2 AR and no MA parameters. Second, the dependent variable is differenced applying the estimated value of *d* and an ARMA model is fit. Third, all parameters (including *d*) are re-estimated with the R package *fracdiff*.

[7] The first two type of models are discussed in Franses and van Dijk (2000, chapter 3) in the context of return series. Case, Guidolin, and Yildirim (2012) provide a recent and very detailed application of univariate and multivariate Markov switching models to REIT returns.

the variance σ_ε^2 do not change between the two regimes, while the other four parameters (ϕ_0, ϕ_1, ϕ_2, ϕ_3) do. These other parameters identify, respectively, a constant, a linear trend and two terms that capture seasonal variation.

The model of Eq. (2) condenses to a second-order Hamilton Markov switching model if the trend and seasonal terms are moved out of the second parenthesis term and to the right-hand side of the equation. In practice this causes convergence problems, which is why we do not present the Hamilton version of the switching model.

In the smooth transition model the return series behaves differently depending on the observed housing price return in the previous period. The transition from one state or regime to another is assumed to occur over a number of time periods rather than abruptly.[8] The particular model we employ can be written as

$$\begin{aligned}(1-\rho L)(1-L)[y_t-(1-G)(\alpha_1 \sin .t+\alpha_2 \cos .t)\\ -G(\beta_1 \sin .t+\beta_2 \cos .t)]=(1+\theta)\varepsilon_t,\end{aligned} \tag{3}$$

with $\varepsilon_t \sim N\left(0, \sigma_\varepsilon^2\right)$ and where

$$G=[1+\exp(-\gamma(y_{t-1}-c))]^{-1}. \tag{4}$$

The model differs from the Markov switching model of Eq. (2) in that a unit root is imposed and, instead of two AR parameters, there is a combination of an AR and an MA parameter in Eq. (3). The smooth transition model of Eq. (3) is similar to the Markov switching model of Eq. (2) in that the coefficients of the seasonal component are switching between regimes; but there is no constant or trend term present. The switching between regimes is tied to the multiplier G, which is itself a logistic function of the dependent variable at lag 1. The lag of the dependent variable is also known as the transition variable. Its unknown critical value is c. The adjustment parameter γ determines the speed of adjustment from one regime to the other. Large values suggest fast adjustment, small values slow adjustment. G takes on values between 0 and 1. If G is equal to 0, only the model parameters α_i are active. If $G=1$, only the parameters identified as β_i are active; if G lies between 0 and 1, the parameters of the constant, the trend, and the seasonal variables are weighted averages of the α_i and the β_i.

Nonlinear moving average models of the type proposed by Engle and Smith (1999), González (2004), Gonzalo and Martinez (2006) allow for stochastic switching between a regime that displays stationary behavior and one that is non-stationary. This may be a useful modeling approach for housing price returns given that returns tend to rise during a bubble, fall when the bubble bursts, but otherwise are close to stationary. Similar to the smooth transition model, the regime switches are tied to past observed values. These are past innovations or shocks (ϵ_{t-1}) in the case of the nonlinear moving average model. The size of the shock determines which regime prevails, with large shocks driving the stochastic trend and small shocks having only temporary effects.

Engle and Smith (1999) suggest a moving average model of the type

$$\triangle z_t=\epsilon_t-g_{t-1}\epsilon_{t-1}, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2) \tag{5}$$

where z_t is $I(1)$ and where the variable g depends on both the parameter γ and the square of the current shock,

$$g_t=1-\frac{\epsilon_t^2}{\gamma+\epsilon_t^2}, \quad \gamma>0. \tag{6}$$

For large shocks, variable g tends toward 0, which turns Eq. (5) into a random walk; the shock will have a permanent effect. Small shocks force g to approach 1, which makes z approximately stationary. In what follows, the model consisting of Eqs. (5) and (6) will be called the STOPBREAK model.

Gonzalo and Martinez (2003) suggest a modification of Eq. (5) with a threshold parameter r,

$$g_t=\begin{cases}\theta_1, & |\epsilon_t|<r\\ \theta_2, & |\epsilon_t|\geqslant r.\end{cases} \tag{7}$$

This modification converts the model of Engle and Smith (1999) from a one-parameter into a three-parameter model.[9] This model variation will be identified as STIMA in the remainder of the paper.

To estimate Eq. (5), we assume $y_t=\triangle z_t$ and also allow for autocorrelation and seasonal variation. With these modifications, our model can be written as

$$\begin{aligned}&(1-\rho L)(y_t-\beta_0-\beta_1 \sin .t-\beta_2 \cos .t)\\ &\quad=\epsilon_t-g_{t-1}\epsilon_{t-1}, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2),\end{aligned} \tag{8}$$

where g_t is specified according to either model variants 6 or 7.[10]

2.3. Structural time series models

Our third model category is based on structural time series models, which are also known as unobserved component models or state space models.[11] Models in this class decompose the series to be forecast into a number of unknown but estimable components, such as trend, cycle, or season. Unlike classical decomposition methods, each of the unobserved components can be stochastic, which implies that the features of the individual components may change over time.

The state space model consists of an observation equation and a state equation. For a standard structural time series model, the observation equation contains a trend term μ_t and an error term ε_t,

$$y_t=\mu_t+\varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2). \tag{9}$$

The corresponding state equation specifies the trend term μ_t. A random walk is a common specification for the trend,

$$\mu_t=\mu_{t-1}+\eta_t, \quad \eta_t \sim N\left(0, \sigma_\eta^2\right). \tag{10}$$

The model consisting of Eqs. (9) and (10) is known in the structural time series literature as a local level model. Both the trend term

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[8] The smooth transition model is discussed in detail in Franses and van Dijk (2000)) and in Teräsvirta (1994, 2004). A recent application to the real estate market is provided in Füss, Stein, and Zietz (2012).

[9] The model introduced by Gonzales (2004) adds yet another two parameters and is, therefore, even more flexible. However, the added flexibility comes at the price of added complexity, which shows up in significant convergence problems for the data set at hand. As the model fails to converge for a number of subsets of the data, we do not further discuss it.

[10] All models in this category are estimated with the program package *TSM* (Time Series Modelling), version 4.39, by James Davidson (http://www.timeseriesmodelling.com/). Open source alternatives are available, such as the *R* package *tsDyn* or the stand-alone program JMULTI for the smooth transition model. Estimation of Markov switching models is discussed at http://weber.ucsd.edu/jhamilto/software.htm#Markov.

[11] Commandeur and Koopman (2007) give a nontechnical overview of the methodology. More detail is available in Harvey (1989) and in Durbin and Koopman (2001). There is a growing number of applications in the housing economics literature, such as Chen (2003), Chen and Sing (2006), and Clark and Coggin (2009).

μ_t and the error term ε_t are unobserved components of the model, each with an unknown variance. Estimation of the model means identifying the values of the two (hyper-)parameters that tie down the model statistically, the variances σ_ε^2 and σ_η^2.

The basic model consisting of Eqs. (9) and (10) can be extended by adding components. A standard extension for variables containing seasonal elements is the inclusion of a seasonal component (S_t) in the observation equation,

$$y_t = \mu_t + S_t + \varepsilon_t, \quad \varepsilon_t \sim N\left(0, \sigma_\varepsilon^2\right), \tag{11}$$

which implies for monthly data the addition of the state equation,

$$S_t = -S_{t-1} - S_{t-2} - \cdots - S_{t-11} + \nu_t, \quad \nu_t \sim N(0, \sigma_\nu^2) \tag{12}$$

or an equivalent equation consisting of trigonometric terms. The inclusion of an error term ν_t in the state equation for the seasonal component allows the component to vary over time. Leaving off the error term forces the seasonal component to be fixed in its influence over time.

In the long run, the evolution of y_t is driven by the stochastic trend μ_t. By modifying Eq. (10) it is possible to allow for a growth term in the stochastic trend. This is of particular relevance for the present study where a boom/bust type situation is modeled. A standard method to allow for a growth term either in the upward or downward direction is to include a drift or slope parameter β in the state equation for μ_t,

$$\mu_t = \mu_{t-1} + \beta_{t-1} + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2). \tag{13}$$

The drift or slope parameter itself can be modeled as a random walk,

$$\beta_t = \beta_{t-1} + \zeta_t, \quad \zeta_t \sim N(0, \sigma_\zeta^2). \tag{14}$$

The model consisting of Eqs. (9), (13) and (14) is known as a local linear trend model, with three hyper-parameters to be estimated. In standard Box-Jenkins ARIMA terms, the addition of Eq. (14) makes y_t into an $I(2)$ variable.

By including a seasonal component and combining Eqs. (11)–(14) into a single model one obtains the basic structural model with four hyper-parameters. More complicated models arise if additional components are added, such as an autoregressive component, which can capture short-run deviations from a trend, or a cyclical component, which can approximate long-run deviations from a trend. Although of value in other applications, none of these extensions turn out to be useful additions in the current context of house price returns.

The models summarized so far can be modified by restricting some of the hyper-parameters. For example, by forcing the variance of the level Eq. (13) to zero ($\sigma_\eta^2 = 0$) one can generate a smooth stochastic trend, where all stochastic variation in the trend arises from the disturbance of Eq. (14), that is, it comes in the form of a shock to the slope of the trend rather than from a shock to its level. In the case of the basic structural model, the error term of Eq. (11) is sometimes set to zero ($\sigma_\varepsilon^2 = 0$) to help in the identification of the three other hyper-parameters.[12]

Hyndman et al. (2008) suggest a number of additional modifications of the structural time series models to enhance their forecasting performance. Their ETS model, for example, which stands for error-trend-seasonal, modifies the error structure of three equations of the basic structural model in that all three equations are driven by the same error term, with the provision that the errors of the level and slope equations are scaled versions of the error of the observation equation. The ETS model also allows for a number of variations in each of its components. In particular, five different versions of the trend model are considered (none, additive, additive damped, multiplicative, and multiplicative damped), and each one of them can be combined in principle with one of three seasonal component choices (none, additive, or multiplicative). This allows for a sizable number of alternative models with multiple parameter restrictions that can in principle behave rather differently.[13] The BATS group of models, as discussed in De Livera, Hyndman, and Snyder (2011), is a further elaboration of the ETS model family. It allows for Box–Cox transformations of the dependent variable, ARMA errors for the observation equation, a damping factor for the slope, and a more complex seasonal structure.[14]

In this context it is noteworthy that standard structural time series models can be transformed after estimation into an equivalent ARIMA format. For example, the model consisting of Eqs. (9) and (10) can be shown to be equivalent to a standard ARIMA(0, 1, 1) model.[15] However, the mathematical equivalence that can be established expost does not mean that both methods are equally good at forecasting. This point has been addressed in some detail, inter alia, by Harvey (1997) and Commandeur and Koopman (2007). The key message can be summarized as follows. Trends in the mean or the season are treated as nuisance components in ARIMA modeling and effectively removed from the modeling process by differencing. This requires pre-testing for stationarity. But stationarity tests tend to have low statistical power and are not good at detecting changes in the underlying trend behavior. In structural time series models, no pretesting is required. All model components, including the underlying trend, are estimated simultaneously and as part of the model. All components can be specified to be stochastic, which allows them to change over time. As change plays a key role for forecasting in economics and finance, being able to incorporate change explicitly in the underlying trend or the seasonal component is a significant advantage to more traditional forecasting methods, at least in theory. Whether this also translates into better forecasting in practice is an empirical question that is addressed in the following section in the context of forecasting the housing price downturn.

3. Empirical results

3.1. Summary of estimated models and their in-sample fit

Table 1 provides a summary of the key characteristics and of the in-sample fits of the models considered in this study. Statistical fit is measured with root mean squared error (RMSE). Two in-sample RMSE values are provided. The first one covers the sample up to the first forecast point, December 2004, when the data provided little evidence of an impending housing price downturn. The sample for the second RMSE value extends up to the last forecast point, June 2007, when housing prices had already moderately declined for a year. The neural network and the naive models provide, respectively, the best and worst in-sample fits for either sample period.

12 Estimation of the basic structural time series models discussed above is done with STAMP 8 (Koopman, Harvey, Doornik, & Shephard, 2007). Equivalent estimates can be obtained by the UCM procedures of SAS/ETS or STATA. Commandeur, Koopman, and Ooms (2011) provide a survey of statistical packages that can estimate structural time series models.

13 The *ets* function in the *forecast* package of *R* allows for an automatic search over many of these model variants and has performed rather well in forecasting competitions (Hyndman, Koehler, Snyder, & Grose, 2002).

14 These variations of the ETS family are implemented in the function *bats* of the *forecast* package of *R*.

15 See Harvey (1989, Appendix 1, pp. 510–511) for a tabular survey of the equivalence of standard structural time series models and ARIMA models.

Table 1
In-sample fit and key characteristics of forecasting methods.

Method	RMSE		Key model characteristics	
	to 12/04	to 06/07	to 12/04	to 06/07
ARIMA-style				
ARIMA	0.176	0.183	$(2, 1, 3) \times (2, 0, 1)_{12}$	$(1, 1, 3) \times (1, 0, 1)_{12}$
ARFIMA	0.207	0.191	(3, 0.493, 2)	(4, 0.499, 5)
Switching				
Markov	0.142	0.150	$P(1\|1) = 0.75$ $P(2\|2) = 0.94$	$P(1\|1) = 0.73$ $P(2\|2) = 0.93$
Smooth transit.	0.186	0.190	$\gamma = 13.1$, $c = 1.58$	$\gamma = 12.1$, $c = 1.53$
STOPBREAK	0.190	0.193	$\gamma = 293.7$	$\gamma = 4170$
STIMA	0.186	0.193	$\theta_1 = 0.009$, $\theta_2 = -0.829$, $r = 0.518$	$\theta_1 = -0.601$, $\theta_2 = -0.050$, $r = 0.000$
Struct. time series				
Local level	0.182	0.189	$\sigma_\varepsilon^2 = 0, \sigma_\eta^2 = 0.027$	$\sigma_\varepsilon^2 = 0, \sigma_\eta^2 = 0.036$
Local trend	0.182	0.189	$\sigma_\zeta^2 = \sigma_\varepsilon^2 = 0, \sigma_\eta^2 = 0.028$	$\sigma_\zeta^2 = \sigma_\varepsilon^2 = 0, \sigma_\eta^2 = 0.029$
Smooth trend	0.212	0.216	$\sigma_\zeta^2 = 2.03\sigma_\varepsilon^2, \sigma_\varepsilon^2 = 0.006$	$\sigma_\zeta^2 = 1.99\sigma_\varepsilon^2, \sigma_\varepsilon^2 = 0.006$
ETS	0.179	0.186	(A,N,A)	(A,N,A)
BATS	0.178	0.179	(1, {0,0}, 0.981, {12})	(1, {4,0}, 0.995, {12})
Other				
Naive	0.256	0.251	(0,1,0)	(0,1,0)
Neural net	0.115	0.129	13-8-1, 121 weights	13-8-1, 121 weights

Notes: All models are estimated on monthly seasonally unadjusted data starting with October 1987. All models contain a seasonal component, except for the naive, neural net, and arfima models. The naive model is based on a ARIMA(0,1,0) random walk.

This is not surprising. The neural network uses a total of 121 parameters to fit either model, which is far larger than the number of parameters estimated for all other models combined. By contrast, the naive method uses no estimated parameters.

The Markov switching model has the second best fit statistics for either sample period. This supports the recent experience of Case et al. (2012), who find the Markov approach to yield superior in-sample fits relative to standard alternative models.[16]The probabilities of the Markov transition matrix do not change much between our first and last forecasting points. The probability of remaining in regime 1 instead of moving into regime 2 is around 75%, while the probability of staying in regime 2 rather than switching to regime 1 in the next period is around 94%. In either case, the switch probabilities are rather low, but not negligible.

The gap in terms of RMSE between the Markov model and the method with the next best in-sample fit is rather large. Perhaps as surprising is the fact that the venerable Box–Jenkins ARIMA model fares better than the BATS model for the sample period up to December 2004 and is closer to the BATS model than any of the other models for the sample up to the forecasting point of June 2007. What contributes to the good in-sample fit of the ARIMA models is their relative complexity, with 8 autoregressive (AR) and moving average (MA) parameters in the model for the first sample period. This is at the upper end of ARIMA model complexity that would ordinarily be considered in practice. We note in this context that the in-sample fit of the ARFIMA model is considerably worse, although the number of estimated AR and MA parameters is comparable to that of the ARIMA model. The likely reason for the worse fit is the absence of explicit seasonal parameters. An interesting aspect of the estimation results for the ARFIMA model is the value of the fractial difference parameter *d*. Its value of close to 0.5 supports the idea that the monthly return series that is used as the dependent variable cannot easily be determined as either stationary or in need of first differences. As mentioned, this ambiguity is borne out by the results of typical tests for unit root behavior: different conclusions emerge dependent on the time frame considered and the test employed.

With the exception of the Markov model, the switching regression models perform less well in-sample than the structural time series models, again with one exception, the smooth trend model, which has the worst in-sample fit of all the models apart from the naive model. The smooth transition model has the second best in-sample fit among the switching models. Its parameters are rather stable across the first and last sample period. This includes the critical value (c) at which a switch is likely to happen for the return series. The adjustment parameter γ is small enough to suggest that the transition to a new regime is smooth rather than instantaneous. Stability in the key parameters is not a characteristic of either the STOPBREAK or the STIMA versions of the nonlinear moving average model. Both models experience dramatic parameter changes between the first and last estimation sample, which indicate a strong model response to the change in the return series around 2006.

The local level and the local linear trend models of the structural time series class of models perform almost identically in-sample. This is not surprising as most of the variation originates with the error term of the level, which is Eq. (10) for the level model and Eq. (13) for the trend model. In both models, the variance of the error term of the observation equation (σ_ε^2) is estimated to be zero. The variance of slope Eq. (14), (σ_ζ^2), is also estimated to be zero, which makes the slope non-stochastic, although not necessarily equal to zero.[17]

The smooth trend model forces the variation in the level equation to zero, which means that the variances of the slope and the observation equations serve as shock absorbers. As a result, the variance of the error term of the slope equation is relatively large, close to half the variance estimated for the error term of the level equation of the local trend model. The practical consequence of a large variance in the slope equation is a strong response to changes in the trend. In other words, the equation responds sensitively to

[16] Case et al. (2012) focus on the return series of REIT, stock and bond indices, which tend to be more volatile than the housing price index considered in this study. As a consequence, their basis of comparison are GARCH style models, which are standard models for returns with considerable volatility.

[17] It should be noted in this context that each of the first three structural time series models of Table 1 contains a stochastic seasonal component with a small variance. Therefore, the local trend model identified in Table 1 is equivalent to what Harvey (1989) calls the basic structural model.

turning points. Ideally, this could make for a good turning point indicator. On the negative side, it could lead to too many false alarms of a turning trend.

The ETS models are identified in Table 1 as consisting of the three terms *A*, *N*, *A*, which need to be interpreted as an additive error term (*A*), no trend/slope parameter (*N*), and an additive seasonal component (*N*). Apart from the restrictions on the variances of the error terms, the ETS model is similar to the estimated local trend model because the variance of the slope is estimated to be zero in both cases. The BATS model, which adds several potential parameters to the ETS model, contains no transformation of the Box–Cox type, as identified by the parameter of unity, and has a damping factor slightly less than unity for the slope. The model for the forecasting point 12/04 shows no ARMA errors for the observation equation, while the model for the last forecasting point (06/07) contains four AR but no MA parameters (4, 0).

The information presented in Table 1 is not informative about the process of model discovery. In practice, this is of importance because well fitting models that are difficult to estimate or even more difficult to discover may not be very useful for routine forecasting tasks. The models presented in Table 1 can be put into three categories in terms of model discovery: (a) automated model discovery based on best-practice routines, (b) manual discovery with accepted rules and a limited number of model variants, and (c) manual discovery with few or no accepted rules and many modeling options. Of the models described in Table 1, ARIMA, ARFIMA, ETS, BATS, and neural network models belong to the first category.[18] Structural time series models outside the specialized ETS and BATS implementations belong into the second category. Switching models are in the third category, which makes estimation and forecasting an order of magnitude more difficult.[19] Given the ambiguity of model discovery with switching models, we cannot exclude the possibility that one can identify switching models that outperform the ones used in this paper. This possibility can be excluded with some degree of confidence for the other model classes.

3.2. Quantitative assessment of forecasting performance

We divide the assessment of the out-of-sample forecasting results into two parts, (a) a quantitative evaluation on the basis of the same measure of out-of-sample forecasting fit that we use for in-sample fit (RMSE) and (b) a qualitative, mainly graphical evaluation of the forecasting methods based on their ability to provide early warning signs of a downturn in housing prices.

Tables 2–4 summarize the out-of-sample forecasting performance in terms of RMSE at 6, 12, and 18 months past the end of the estimation period. All samples start at the same time in 1987. They end at the forecast points identified in Tables 2–4. The forecast points are six months apart starting with December of 2004. For the naive model, which can be thought of as a simple benchmark, the value of the house price return at the forecast point is assumed to prevail without change month after month for the next 6, 12, or 18 months into the future.

Table 2 shows the forecasting results for the naive model, which has the worst in-sample fit, the neural network model with the best in-sample fit, and the two ARIMA-type models. The neural network with its large number of coefficients noticeably outperforms the naive model in terms of RMSE for the first two forecast points,

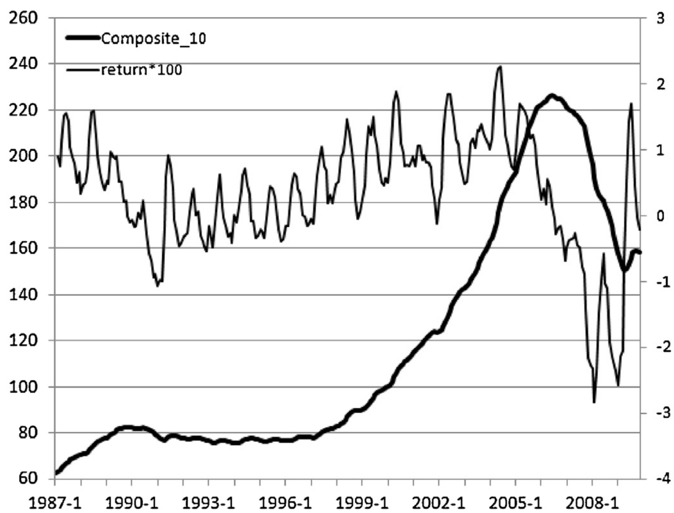

Fig. 1. Case-Shiller price index and corresponding return series.

December 2004 and June 2005. For a forecast horizon up to the middle of 2006, the return series stays within a range of values that is consistent with those of the estimation sample. Therefore, a model with an exhaustive number of parameters, such as the neural network model, fits relatively well. However, once the return series takes on values not seen in the estimation sample, the forecasts of the neural network model are considerably worse than those of the naive model. These results are consistent with general forecasting experience: over-fitted models typically do not forecast well out-of-sample if there are noticeable changes in the series that needs forecasting.

The naive model stands out in terms of forecasting accuracy at the forecast point December 2005. The forecast horizon at this point covers the months when the return series falls sharply (Fig. 1). The naive model, with its prediction of little change, is outperformed at this critical point only by the smooth trend model (Table 4). None of the other methods comes close in terms of accuracy.

The ARIMA model performs very well in stable environments, which covers a forecast horizon through December 2005 in Table 2. It is outperformed by most other forecasting methods when there is sizable change in the direction and magnitude of the dependent variable. The ARFIMA model does not differ dramatically from the ARIMA model in this respect. However, on average, as reported in the last line of Table 2, it performs worse than the ARIMA model. In fact, it is worse than the naive model in average forecasting performance.

The Markov switching model, which has an exemplary in-sample fit according to Table 1, performs below average in the out-of-sample forecasts (Table 3). At all forecast points it is outperformed by at least one of the other switching regression models and typically also by the structural time series models of Table 4. Except for the forecast point December 2005, the smooth transition model has a consistently better forecasting accuracy than the Markov model. For the December 2005 forecast point, however, the two nonlinear moving average models tend to dominate the Markov model. In fact, the STOPBREAK and STIMA models outperform the Markov model consistently except for the last two forecast points, December 2006 and June 2007. But even for those points they do well relative to the Markov model at a 6 months time horizon. The average RMSE values at the end of Table 3 rank the smooth transition model first among the switching models, followed closely by the nonlinear moving average models, with the Markov model being last. Comparing the RMSE averages at the end of Table 3 to those at the end of Table 2, it is apparent that

[18] All of them can be estimated with the *forecast* package of *R*.

[19] Helpful procedures for model discovery of smooth transition models are available in the stand-alone package JMULTI (http://www.jmulti.de/) and in the *R* package *tsDyn*.

Table 2
Out-of-sample forecasting performance (RMSE) of naive, neural network, and ARIMA-style models.

Forecast point Forecast horizon	Naive	Neural network	ARIMA	ARFIMA
December-04				
January-05 to June-05	0.788	0.436	0.257	0.436
January-05 to December-05	0.631	0.397	0.277	0.342
January-05 to June-06	0.566	0.527	0.743	0.465
June-05				
July-05 to December-05	0.562	0.508	0.255	0.225
July-05 to June-06	0.924	0.682	0.877	0.584
July-05 to December-06	1.282	1.234	1.166	0.888
December-05				
January-06 to June-06	0.230	0.855	0.897	0.507
January-06 to December-06	0.597	1.284	1.129	0.975
January-06 to June-07	0.704	1.283	1.314	1.070
June-06				
July-06 to December-06	0.469	1.415	0.285	0.765
July-06 to June-07	0.494	1.146	0.405	0.939
July-06 to December-07	1.016	1.657	0.816	1.427
December-06				
January-07 to June-07	0.322	0.732	0.238	0.701
January-07 to December-07	0.698	1.147	0.730	1.193
January-07 to June-08	0.961	1.568	1.035	1.696
June-07				
July-07 to December-07	1.118	1.314	0.831	1.021
July-07 to June-08	1.349	1.483	1.092	1.606
July-07 to December-08	1.384	1.603	1.038	1.647
Overall average	0.783	1.071	0.744	0.916

Notes: All models are estimated on monthly seasonally unadjusted data starting with October 1987 and ending with the forecast point. The forecast horizon extends 6, 12, and 18 months into the future from the forecast point.

the Markov model performs less well than the naive model, while the other three switching models edge out all models reported in Table 2.

On average, the forecasting performance of the structural time series models (Table 4, last line) is better than that of the models reported in Tables 2 and 3. In terms of the simple average of all reported RMSE values, the BATS model outperforms all other structural time series models. The smooth trend model performs the worst among the structural time series models. But its average RMSE is still lower than that of any model reported in Tables 2 or 3 other than that of the smooth transition model. The relatively high average RMSE value of the smooth trend model results from the model's much worse than average performance in June 2005 and in June 2006. At the critical forecast point of December 2005

Table 3
Out-of-sample forecasting performance (RMSE) of switching models

Forecast point	Markov	Smooth	Nonlinear moving average	
Forecast horizon	Switching	Transition	STOPBREAK	STIMA
December-04				
January-05 to June-05	0.898	0.320	0.332	0.384
January-05 to December-05	0.673	0.269	0.300	0.369
January-05 to June-06	0.725	0.664	0.467	0.429
June-05				
July-05 to December-05	0.488	0.278	0.247	0.253
July-05 to June-06	0.703	0.619	0.531	0.487
July-05 to December-06	0.915	0.792	0.722	0.636
December-05				
January-06 to June-06	0.704	0.835	0.764	0.679
January-06 to December-06	0.907	0.981	0.906	0.809
January-06 to June-07	1.158	1.202	1.042	0.925
June-06				
July-06 to December-06	0.570	0.212	0.190	0.185
July-06 to June-07	0.574	0.287	0.383	0.350
July-06 to December-07	0.978	0.651	0.837	0.786
December-06				
January-07 to June-07	0.793	0.318	0.427	0.479
January-07 to December-07	0.978	0.802	0.939	1.016
January-07 to June-08	1.402	1.199	1.375	1.474
June-07				
July-07 to December-07	0.734	0.541	0.704	0.747
July-07 to June-08	0.929	0.851	1.149	1.229
July-07 to December-08	0.934	0.851	1.248	1.347
Overall average	0.837	0.648	0.698	0.699

Notes: All models are estimated on monthly seasonally unadjusted data starting with October 1987 and ending with the forecast point. The forecast horizon extends 6, 12, and 18 months into the future from the forecast point.

Table 4
Out-of-sample forecasting performance (RMSE) of structural time series models.

Forecast point Forecast horizon	Local level	Local trend	Smooth trend	ETS	BATS
December-04					
January-05 to June-05	0.318	0.319	0.301	0.344	0.387
January-05 to December-05	0.246	0.246	0.292	0.272	0.299
January-05 to June-06	0.608	0.608	0.819	0.603	0.587
June-05					
July-05 to December-05	0.177	0.178	0.763	0.268	0.197
July-05 to June-06	0.645	0.641	0.763	0.613	0.640
July-05 to December-06	0.869	0.864	1.019	0.812	0.884
December-05					
January-06 to June-06	0.868	0.864	0.436	0.819	0.753
January-06 to December-06	1.031	1.026	0.385	0.985	0.920
January-06 to June-07	1.218	1.210	0.386	1.172	1.096
June-06					
July-06 to December-06	0.188	0.208	1.004	0.256	0.381
July-06 to June-07	0.286	0.269	1.526	0.279	0.370
July-06 to December-07	0.641	0.587	1.914	0.606	0.699
December-06					
January-07 to June-07	0.393	0.372	0.176	0.344	0.364
January-07 to December-07	0.821	0.780	0.336	0.750	0.586
January-07 to June-08	1.203	1.144	0.540	1.131	0.898
June-07					
July-07 to December-07	0.599	0.570	0.284	0.512	0.506
July-07 to June-08	0.940	0.891	0.548	0.822	0.766
July-07 to December-08	0.932	0.860	0.686	0.806	0.764
Overall average	0.666	0.647	0.677	0.633	0.617

Notes: All models are estimated on monthly seasonally unadjusted data starting with October 1987 and ending with the forecast point. The forecast horizon extends 6, 12, and 18 months into the future from the forecast point.

and at the beginning of the downturn in housing prices (forecast points December 2006 and June 2007), it has the best forecasting performance of any model considered.

3.3. *Qualitative assessment of the ability to predict the housing downturn*

The qualitative assessment considers the ability of the various forecasting methods to forewarn investors of the downturn in housing prices. For this purpose we convert the return forecasts, which are at the basis of Tables 2–4, into forecasts of the actual price index. The resulting forecasts are presented in graph form for a number of representative models (Figs. 2–7). In each of the graphs, the actual price index is shown along with 18-months-ahead forecasts from the six forecast points that are used in Tables 2–4.

For the case of the downturn in U.S. housing prices around 2006 two forecasting criteria appear to be of key importance. First, the forecast should forewarn investors of a downturn in housing prices as early as possible. Second, a good forecasting method should leave investors with no doubt that the downturn in housing prices is substantial once it starts.

The forecasts of the ARIMA model in Fig. 2 fail to provide any evidence of a downturn in housing prices for all forecast points through December 2005. On the contrary, the model forecasts a further increase in housing prices. This prediction failure is not unique to the ARIMA model. All models except for the smooth trend model (Fig. 6) share this result, although the models differ somewhat in the degree to which future housing prices are overestimated at the critical forecast point December 2005. For example, the ARIMA model over-predicts more than the Markov model or the

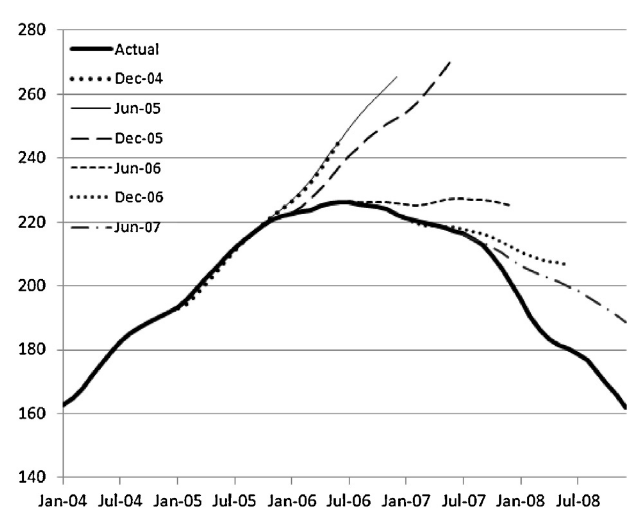

Fig. 2. Forecasts of ARIMA model.

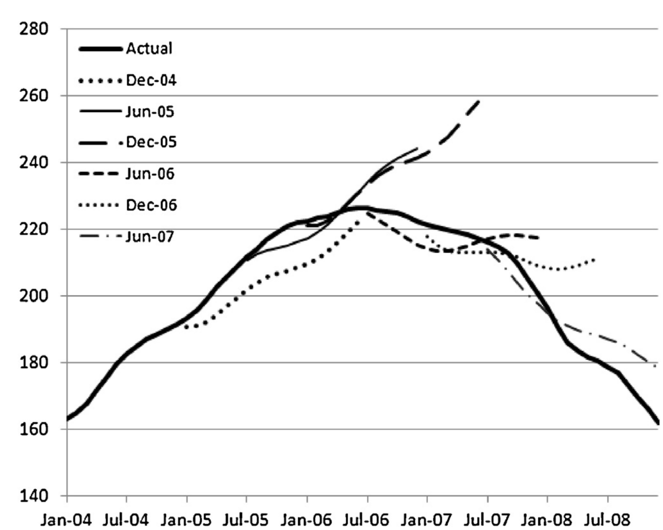

Fig. 3. Forecasts of Markov switching model.

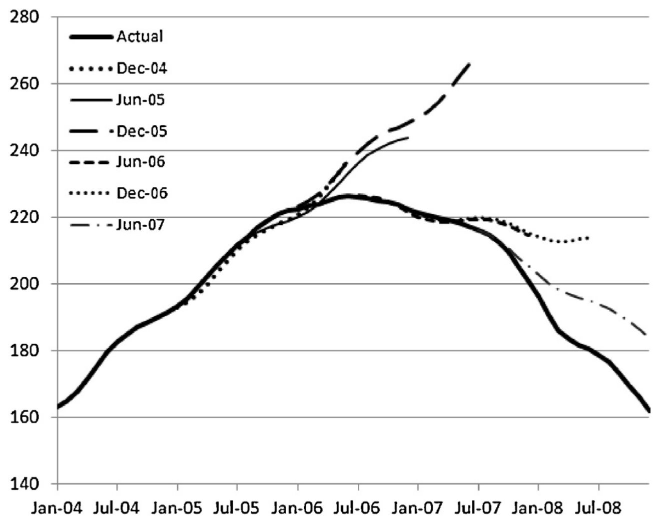

Fig. 4. Forecasts of smooth transition model.

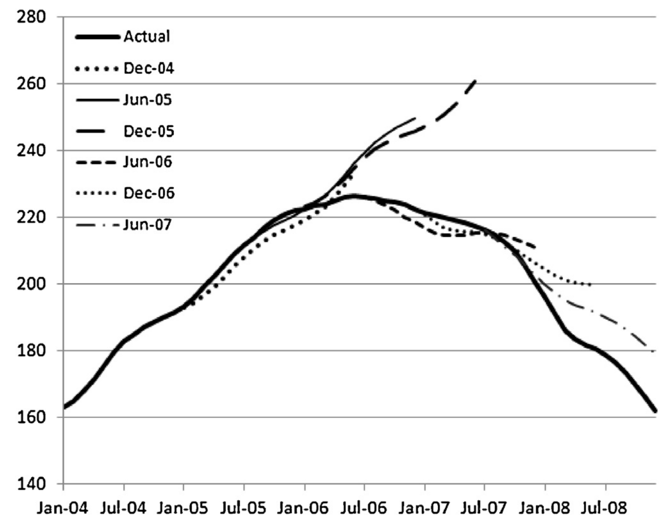

Fig. 7. Forecasts of BATS model.

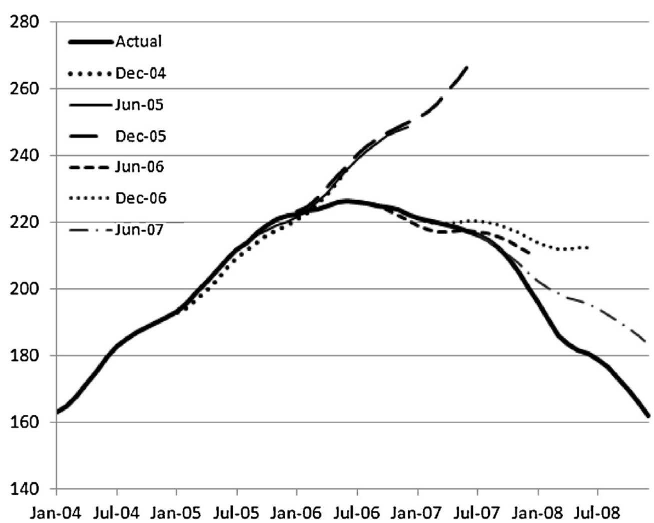

Fig. 5. Forecasts of local linear trend model.

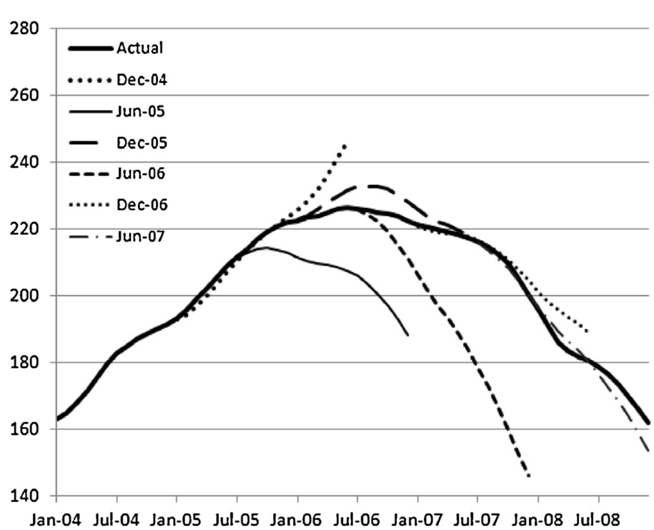

Fig. 6. Forecasts of smooth trend model.

BATS model. However, in terms of the ability to forewarn investors of a housing price downturn as early as possible, only the smooth trend model delivers a good result (Fig. 6). It clearly points toward a downturn at the end of 2005 and it predicts at that time almost perfectly the average housing price level 18 months ahead. Most other forecasting models identify a downward trend only at the forecast point June 2006. The ARIMA model (Fig. 2) arrives at this forecasting result only at the end of 2006. It is, therefore, not surprising that housing market analysts relying on such predictions were unable to see any evidence of a looming downturn in the housing market, but rather considered a soft landing a reasonable prediction throughout much of 2006.

The ability of the smooth trend model to provide a better forecast than any other model at the critical forecast point December 2005 is known from Table 4. Not apparent from Table 4 is the reason behind the model's inferior forecasting performance half a year earlier (June 2005) and half a year later (June 2006). Fig. 6 provides the reason: the two instances of below average forecasting performance of the smooth trend model are associated with predicting the coming downturn in prices about a year too early. In fact, the smooth trend model consistently predicts a downturn for all models starting with the forecast point June 2005. Only the very first forecast in December 2004 is silent about a future downturn. From this evidence we conclude that the high RMSE values of the smooth trend model at the forecast points June 2005 and June 2006 do not detract from the model's excellent forecasting performance in the wake of the housing price downturn. Investors would have been forewarned with ample lead time of a pending downturn in home prices. There is no other model of similar value to investors.

The smooth trend model also excels when it comes to predicting the severity of the downturn in prices, once prices are turning down in the middle of 2006. As already evident from the results of Table 4, the model's forecasting performance is far better than that of any other model at this point. The BATS model (Fig. 7) comes in second and, at the very last forecasting point (June 2007), the Markov model (Fig. 3) also performs well.

When looked at from the perspective of providing both an early warning of an impending downturn in prices and a sense of the severity of the downturn once prices are falling, the smooth trend model performs best. On the downside, the model predicts the downturn a year too early in two of five cases. However, by many investors this may actually be considered a positive rather than a negative model characteristic. What is truly noteworthy in this context is that the most useful forecasting technique, the smooth trend model, has a worse in-sample fit than all models other than the naive model. By contrast, the model with the best in-sample fit, the neural network model, turns in the worst forecasting performance, worse than that of the naive model. We conclude that good in-sample fit is no guarantee for good out-of-sample forecasting. This well-known result applies a fortiori to the prediction of the downturn in U.S. housing prices that started around 2006.

4. Conclusions

The study analyzes to what extent univariate time series models could have provided investors with useful forecasts of housing prices before or during the early stages of the downturn in U.S. housing prices around 2006. Two questions are considered: (a) which if any methods could have provided sufficient lead time to react to an impending housing price downturn? (b) given that housing prices started to decline in 2006, which if any methods could have provided a realistic forecast of the actually experienced severity of the downturn?

The study considers three classes of univariate forecasting models, ARIMA-style models, switching regressions, and structural time series models. The dependent variable is defined in terms of the monthly return on the 10-city, seasonally unadjusted, composite index of the Standard and Poor's Case-Shiller Home Price Index. Dynamic, out-of-sample forecasts are provided at 6 different forecast points with time horizons of 6, 12, and 18 months into the future. The forecast points are selected at 6 months intervals starting with December of 2004. In-sample and out-of-sample fits are measured in terms of root-mean-squared-error (RMSE).

The estimation results identify notable differences in in-sample model fit and out-of-sample forecasting performance across models. Traditional ARIMA models and Markov switching models perform very well in-sample. However, their out-of-sample forecasting results are typically worse than those of other switching models or of the class of structural time series models. The good out-of-sample forecasting performance of the structural time series models is likely tied to the fact that the underlying trend of a series is explicitly modeled by these models in terms of layers of random walks, which gives these models the ability to pick up subtle trend changes relatively easily. The only serious model competitor to the class of structural time series models is the switching model known as smooth transition model.

Within the class of structural time series models, models known as ETS and BATS, introduced into the literature by Hyndman et al. (2002) and by De Livera et al. (2011), provide on average the best out-of-sample forecasting performance. This corroborates the results of earlier forecasting competitions (Hyndman et al., 2002). However, a standard smooth trend model offers by far the most useful early warning signs of a downturn in housing prices. It consistently predicts a downturn in housing prices starting in June of 2005. No other model achieves this result. The sensitivity to trend changes of this particular structural time series model results from the fact that changes in the level of the underlying stochastic trend are entirely driven by shocks to the slope or drift of the trend, not by shocks to the level itself. The smooth trend model is also the forecasting method with the best record of pinning down the severity of the downturn once housing prices start to decline in 2006. Yet this model performs rather poorly in terms of in-sample fit. This supports the well-known forecasting rule that excellent in-sample fit does not necessarily translate into good forecasting performance when there are changes to the series not experienced before.

We conclude that some univariate models could have predicted the downturn in U.S. housing prices as early as the middle of 2005. They could also have predicted the severity of the housing downturn. We suggest that structural time series models are the forecasting models with the most promise to achieve reasonable forecasting results. What adds to their appeal is the fact that, at this point in time, they are easy to estimate with standard software packages.

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